

Krishi-Drishti: Carbon Credit Verification Platform for Indian Farmers

Project Report

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Acknowledgement

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Abstract

Krishi-Drishti is a verified carbon credit platform designed to enable Indian farmers to generate, track, and monetize carbon credits through sustainable agricultural practices. By integrating satellite data from Google Earth Engine, mobile-based farmer engagement, and ML-driven verification, the platform creates a trusted ecosystem for climate finance. This report details the problem of agricultural emissions in India, surveys the existing market landscape, and presents a technical architecture for a scalable, transparent verification system.

Contents

Acknowledgement	i
Abstract	ii
1 INTRODUCTION	1
1.1 Project Overview	1
1.2 Problem Statement	1
1.3 Solution Approach	1
1.4 Target Users	1
2 LITERATURE SURVEY	1
2.1 Existing Systems	1
2.2 Proposed System Philosophy	2
2.2.1 Key Differentiators	2
2.3 Technical Innovation	2
2.4 Problem Statement Reiteration	2
2.5 Project Objectives	2
3 SYSTEM REQUIREMENTS	2
3.1 Hardware Requirements	2
3.2 Software Requirements	2
4 SYSTEM DESIGN	3
4.1 Architecture Overview	3
4.2 Verification Process Design	3
4.3 Carbon Credit Calculation	3
4.4 Security Architecture	3
5 IMPLEMENTATION	3
5.1 Development Approach	3
5.2 System Components	4
5.3 Testing Strategy	4
6 CONCLUSION & FUTURE WORK	4
6.1 Project Achievements	4
6.2 Future Enhancement Roadmap	4
6.3 Environmental & Social Impact	4
7 REFERENCES	4

1 INTRODUCTION

1.1 Project Overview

Krishi-Drishti is a verified carbon credit platform designed to enable Indian farmers to generate, track, and monetize carbon credits through sustainable agricultural practices. The platform integrates satellite data, farmer engagement tools, and blockchain-based verification to create a trusted ecosystem for carbon credit generation.

1.2 Problem Statement

Indian agriculture contributes 18% of national GDP and employs 58% of the workforce, but faces two critical challenges:

- **Carbon Emissions Crisis:** Agricultural practices emit 16 – 20% of India's total greenhouse gases through soil degradation, methane emissions, and land use changes.
- **Farmer Income Gap:** Small and marginal farmers (86% of agricultural population) lack access to climate finance and carbon offset income.

Current barriers preventing farmer participation include a lack of standardized verification, complexity in documentation, and non-transparent payment infrastructures.

1.3 Solution Approach

Krishi-Drishti addresses these barriers through:

- **Real Satellite Verification:** Google Earth Engine (Sentinel-2 + SMAP) provides real-time monitoring with 86% confidence scores.
- **Multi-Layer Verification:** Combines satellite data, farmer practice logs, photo evidence, and risk scoring.
- **Farmer-First Design:** Mobile-first interface with Hindi language support and UPI integration.

1.4 Target Users

- **Primary:** Small/marginal farmers (0.5 – 5 hectares).
- **Secondary:** Agricultural cooperatives and carbon aggregators.
- **Tertiary:** Corporate buyers seeking ESG compliance.

2 LITERATURE SURVEY

2.1 Existing Systems

The global carbon market is dominated by registries like Verra and Gold Standard, which have high entry barriers for smallholders. Competitors like **Boomitra** and **EKI Energy** in India provide satellite-based solutions but often rely on manual batch processing or specific aggregator networks.

2.2 Proposed System Philosophy

Krishi-Drishti bridges the gap through a satellite-first approach and an open aggregator network.

2.2.1 Key Differentiators

Table 1: Krishi-Drishti vs. Competitors

Feature	Krishi-Drishti	Boomitra	EKI	Grow Indigo
Tech Stack	Earth Engine + Mobile	Satellite + App	Satellite + Audit	Satellite Only
Verification Speed	7 days	14 days	30 days	60 days
Payment	UPI/Bank Direct	Aggregator-based	Bank Transfer	ACH Only
Risk Scoring	Real-time ML	Batch	Manual	None

2.3 Technical Innovation

Innovation includes **Real-time NDVI Anomaly Detection** to flag unrealistic carbon claims (> 2 standard deviations) and **Multi-Modal Evidence Binding** where photos are geographically and temporally bound to specific farm plots.

2.4 Problem Statement Reiteration

The project addresses the exclusion of 86% of Indian farmers from climate markets due to high transaction costs (15,000 – 50,000 USD per project in traditional models) by reducing the per-farmer verification cost to below 5 USD.

2.5 Project Objectives

The primary objective is to build a platform where a farmer can earn credits within 7 days of registration, ensuring an 80/20 revenue split in favor of the farmer.

3 SYSTEM REQUIREMENTS

3.1 Hardware Requirements

- **Server:** Cloud-based (AWS/GCP), minimum 4 vCPU, 16GB RAM for production.
- **Database:** AWS RDS PostgreSQL with 200GB+ SSD.
- **Client:** Mobile devices with Android 7.0+, GPS, and 5MP+ camera.

3.2 Software Requirements

- **Backend:** Python 3.10+, FastAPI, Google Earth Engine API.
- **Frontend:** React 18.2+, React Native for mobile.
- **Blockchain:** Polygon RPC endpoints (Phase 3 integration).
- **Compliance:** GDPR and India's DPDP Act 2023.

4 SYSTEM DESIGN

4.1 Architecture Overview

The system uses a five-layer architecture: Presentation (Mobile/Web), API (FastAPI), Service (Business Logic), Integration (Satellite/Aggregators), and Data (PostgreSQL/IPFS).

[Diagram Placeholder: System Architecture Diagram]
Insert relevant graphic here.

4.2 Verification Process Design

Verification occurs in five distinct layers:

1. **Layer 1:** Satellite Data (NDVI/SMAP).
2. **Layer 2:** Practice Verification (Photo Geotagging).
3. **Layer 3:** Baseline Comparison (Lab tests + Historical satellite data).
4. **Layer 4:** ML Risk Assessment (Anomaly detection).
5. **Layer 5:** Registry Enrollment.

4.3 Carbon Credit Calculation

The core methodology for credit calculation is:

$$\text{Credits}_{\text{gross}} = (\text{SOC}_{\text{baseline}} - \text{SOC}_{\text{satellite}}) \times \text{Area}_{\text{ha}} \times \text{Factor}_{\text{practice}}$$

A buffer pool deduction of 15% is applied for regulatory compliance:

$$\text{Credits}_{\text{final}} = \text{Credits}_{\text{gross}} \times 0.85 \times (1 - \text{Risk}_{\text{discount}})$$

4.4 Security Architecture

Security is managed via JWT authentication, AES-256 encryption for data at rest, and TLS 1.3 for data in transit.

5 IMPLEMENTATION

5.1 Development Approach

The project follows an Agile development approach using FastAPI for the backend to handle asynchronous satellite data requests from Google Earth Engine efficiently. React Native is used for the mobile application to ensure a consistent experience across Android and iOS.

5.2 System Components

- **Satellite Integration Service:** Wraps Earth Engine API for NDVI and soil moisture analysis.
- **Risk Scorer:** An ML component using Isolation Forests to detect fraudulent practice logs.
- **Payment Module:** Integrated with Razorpay for automated UPI payouts.

5.3 Testing Strategy

Testing includes unit tests for the carbon formula, integration tests for satellite API calls, and E2E Selenium tests for the farmer registration-to-payment journey.

6 CONCLUSION & FUTURE WORK

6.1 Project Achievements

Krishi-Drishti demonstrates that satellite-based verification can achieve 86% confidence, significantly lowering the barrier for smallholder participation. It enables farmers to earn an additional 200 – 2,000 USD annually.

6.2 Future Enhancement Roadmap

- **Phase 2:** Enterprise B2B portal for corporate ESG buyers.
- **Phase 3:** Blockchain-based immutable credit tracking on Polygon.
- **Phase 4:** Expansion into South Asia and Sub-Saharan Africa.

6.3 Environmental & Social Impact

The platform incentivizes regenerative agriculture, leading to increased soil health and significant carbon sequestration, contributing to India's NDC goals under the Paris Agreement.

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